Exploratory Data Analysis (EDA) Summary Report Template

1. Introduction

Purpose

The purpose of this report is to document the initial **Exploratory Data Analysis (EDA)** conducted on the Geldium Finance credit card dataset. This analysis serves as the foundational step in the broader initiative to reduce high credit card delinquency rates by moving the company from a reactive stance to a proactive, data-driven "Early Warning System." By auditing the dataset's structure and completeness, we ensure that any future AI/ML models are built on a reliable, unbiased, and high-quality foundation.

Goals

- **Assess Data Integrity:** To identify and document missing values, inconsistent data points, and anomalies that could potentially compromise the accuracy of delinquency predictions.
- **Identify Early Risk Indicators:** To uncover patterns and correlations between key variables—such as credit utilization and repayment history—that signal a customer's likelihood of transitioning from **DPD** (Days Past Due) to **NPL** (Non-Performing Loan) status.
- **Leverage GenAI for Insights:** To utilize Generative AI tools to efficiently summarize complex data trends and suggest industry-standard strategies for data treatment, all while maintaining strict data confidentiality.
- **Inform Stakeholder Strategy:** To provide the Head of Collections and the Analytics team with a clear, actionable summary of findings that will shape the design of future predictive modeling and personalized intervention strategies.

2. Dataset Overview

This section summarizes the dataset structure and provides an initial health check on the data quality. The dataset contains a mix of demographic, financial, and behavioral attributes used to track customer delinquency over a 6-month period.

Key Dataset Attributes

- **Number of Records:** 500 entries.
- **Key Variables:**
    - **Customer_ID:** Unique identifier for each account.
    - **Age:** Age of the primary cardholder (Range: 18 to 74 years).
    - **Income:** Monthly income of the customer.
    - **Credit_Score:** Numerical representation of creditworthiness.
    - **Credit_Utilization:** The ratio of the current balance to the credit limit.
    - **Missed_Payments:** Number of payments missed in the historical period.
    - **Delinquent_Account:** Target binary flag (0 = Good, 1 = Delinquent).
    - **Month_1 to Month_6:** Categorical repayment status for the last six months (e.g., "Paid", "Late", "Missed").
- **Data Types:** * **Numerical:** Age, Income, Credit_Score, Credit_Utilization, Missed_Payments, Loan_Balance, Debt_to_Income_Ratio, Account_Tenure.
    - **Categorical:** Customer_ID, Employment_Status, Credit_Card_Type, Location, Month_1 through Month_6.

Initial Data Audit & Anomalies

During the initial review, the following observations were made regarding the reliability of the data:

- **Missing Values:** There are significant gaps in critical financial columns:
    - **Income:** 39 missing records (~7.8% of the data).
    - **Loan_Balance:** 29 missing records.
    - **Credit_Score:** 2 missing records.
- **Duplicates:** No duplicate Customer_ID records were found, ensuring each entry represents a unique account.
- **Logical Inconsistencies:** * **Credit Utilization:** The maximum value is **1.025**, indicating some customers have exceeded their credit limit (over-limit status).
    - **Account Tenure:** Ranges from 0 to 19 years, which appears consistent with a standard credit card portfolio.

3. Missing Data Analysis

Identifying and addressing missing data is critical to ensuring model accuracy. This section outlines missing values in the dataset, the approach taken to handle them, and justifications for the chosen method.

Key Missing Data Findings

- **Variables with Missing Values:**
  - **Income:** 39 missing records (7.8% of the total dataset).
  - **Loan_Balance:** 29 missing records (5.8% of the total dataset).
  - **Credit_Score:** 2 missing records (0.4% of the total dataset).
- **Missing Data Treatment:**
  - **Income & Loan_Balance:** We recommend **Median Imputation**. Specifically, imputing values based on the median of the customer's Employment_Status group to ensure the substituted values are as realistic as possible.
  - **Credit_Score:** Given the very low frequency of missingness (0.4%), **Simple Median Imputation** or **Deletion** of these two specific rows is appropriate.

Justification for Treatment

- **Preventing Bias:** Deleting nearly 8% of the records due to missing Income values could lead to "Listwise Deletion" bias. If lower-income or unemployed individuals are less likely to report income, deleting them would make the portfolio appear "healthier" than it actually is, leading to an underestimation of delinquency risk.
- **Preserving Sample Size:** For a relatively small dataset (500 records), every data point is valuable for training a robust Machine Learning model. Imputation allows us to retain the other 18 variables for those 39 customers.
- **Robustness of Median:** We chose the **Median** over the Mean because financial data like Income and Loan_Balance often contains outliers (as seen in the range up to $199,943). The median provides a "typical" value that is less skewed by extreme high-earners.

4. Key Findings and Risk Indicators

This section identifies trends and patterns that may indicate risk factors for delinquency. Feature relationships and statistical correlations are explored to uncover insights relevant to predictive modeling.

Key Findings

- **Correlations observed between key variables:**
  - **The "Utilization Gap":** Delinquent accounts show a higher average **Credit Utilization (50.7%)** compared to non-delinquent accounts (48.9%). While the linear correlation is modest, it remains a primary driver for risk.
  - **Debt-to-Income (DTI) Pressure:** Delinquent customers have a higher average **DTI ratio (30.6%)** than those who are current. This

suggests that customers whose debt obligations exceed 30% of their income are more likely to struggle with payments.
    - **Non-Linear Risk:** The low individual correlation scores (e.g., Income at 0.04) indicate that delinquency is not caused by a single factor but by a **combination of behaviors**. For example, a high-income customer with high utilization may be just as risky as a low-income customer with many missed payments. This finding directly supports the need for the ML models Charithra has proposed.
- **Unexpected Anomalies:**
    - **Over-Limit Accounts:** We identified **4 customers** with a Credit Utilization ratio greater than **1.0 (100%)**. These individuals are "over-limit" and represent the highest immediate risk of moving into the NPL (Non-Performing Loan) category.
    - **The "Hidden Risk" Segment:** There are **89 customers** who currently maintain a **Credit Score below 400** but are *not yet* flagged as delinquent. These represent a high-priority "Early Warning" group that the Collections team can target proactively before they miss their next payment.
    - **Repayment Inconsistency:** Analysis of the 6-month history shows that even delinquent accounts had "On-time" months, suggesting that delinquency in this portfolio is often intermittent rather than a sudden permanent stop in payments.

5. AI & GenAI Usage

Generative AI tools were integrated into the Exploratory Data Analysis (EDA) workflow to accelerate data cleaning, suggest advanced statistical treatments, and translate technical metrics into actionable business insights. To maintain Geldium's strict data confidentiality standards, no Personally Identifiable Information (PII) was shared with external AI models; instead, metadata, column headers, and high-level summary statistics were used to guide the process.

Applications and Example Prompts

Data Integrity Audit and Schema Analysis GenAI was used to evaluate the data structure by providing the prompt: *"Analyze the schema of a 19-column financial dataset. Identify logical inconsistencies to look for, specifically regarding 'Credit Utilization' and 'Income' for a delinquency study."* This allowed for the identification of "Over-limit" status (Utilization > 1.0) as a critical behavioral risk feature rather than a simple data entry error.

Industry-Standard Imputation Strategy To address the 8% gap in income data, the tool was prompted to: *"Suggest an industry-standard imputation strategy for missing income values in a credit risk dataset, comparing Mean vs. Median vs. Conditional Imputation."* Based on the AI-generated insights, **Conditional Median Imputation** was selected to ensure that imputed values accurately reflected the nuances of different employment categories (e.g., distinguishing between "Self-Employed" and "Full-Time" income levels).

Automated Statistical Code Generation To perform deep-dive analytics, a prompt was issued to: *"Write a Python script using pandas to calculate the correlation between 'Delinquent_Account' and all numeric features, and flag any customer with a credit score below 400 who is not yet delinquent."* This automated the heavy lifting of the coding process, leading directly to the discovery of the **89 "Hidden Risk" customers** who are candidates for early intervention.

Strategic Insight Synthesis Finally, GenAI assisted in bridging the gap between raw data and executive strategy with the prompt: *"Based on a finding that DTI and Utilization are the highest correlates for delinquency, draft a professional summary explaining why linear models might miss complex risk patterns."* The resulting insight justified the shift from broad historical segmentation to the more precise, AI-driven predictive modeling planned for the next stage of the project.

6. Conclusion & Next Steps

Summary of Key Findings

The Exploratory Data Analysis of the Geldium Finance dataset has confirmed that while the portfolio is generally stable, there are significant "hidden" risks that traditional segmentation methods likely overlook.

- **Data Readiness:** The dataset of 500 records has been successfully cleaned and standardized. Critical gaps in income and loan balance data have been addressed using conditional median imputation, ensuring a robust foundation for predictive modeling.
- **Primary Risk Drivers:** Credit Utilization and Debt-to-Income (DTI) ratios have emerged as the strongest indicators of financial distress. Notably, customers exceeding 100% utilization (over-limit) represent an immediate risk of transitioning to NPL (Non-Performing Loan) status.
- **Proactive Opportunity:** We have identified a high-priority segment of **89 customers** who possess sub-400 credit scores but have not yet defaulted. This group represents the most significant opportunity for proactive intervention.
- **Complexity of Risk:** The low linear correlation between individual features and delinquency suggests that risk is driven by a combination of factors (e.g., high utilization coupled with a sudden change in payment behavior) rather than any single variable.

Recommended Next Steps

To move Geldium Finance toward an AI-driven "Early Warning System," the following steps are recommended for the next phase of the project:

1. **Advanced Feature Engineering:** Develop "Trended Features" such as three-month moving averages for credit utilization. Capturing the *direction* of debt growth is likely more predictive than a single snapshot.
2. **Model Development:** Transition from EDA to the training of non-linear Machine Learning models (such as **XGBoost** or **Random Forest**). These models are better equipped to handle the complex interactions between variables identified in this report.
3. **Risk Scoring Engine:** Implement a probability scoring system that ranks every customer on a scale of 0 to 100. This will allow the Collections team to prioritize their efforts based on the severity of predicted risk.
4. **GenAI Personalization Pilot:** Use Generative AI to draft and test empathetic outreach scripts tailored to the "Hidden Risk" segment. The goal is to offer flexible payment solutions before a delinquency event occurs, thereby preserving the customer relationship and reducing the bank's capital reserve requirements.

By executing these steps, Geldium Finance will transition from a reactive collections model to a high-precision, proactive risk management strategy.